

Data S1. Archaeological sites description, related to Figure 1A and STAR Methods

Tongwan City Site (统万城) (Contact: Songmei Hu)

Tongwan City was established in northwestern China in 413 CE by Helian Bobo, the founder of the Daxia Kingdom. Due to its strategic location in the heart of the Ordos region, it developed into an important hub for both commercial and military activities within Inner Asia, a role reinforced by its position along the Silk Road. After the collapse of the Daxia Kingdom, Tongwan City continued to be occupied and utilized in subsequent periods². In the present study, a domestic felid sample (FS12) recovered from excavations at the site was radiocarbon dated to 706-883 cal CE, corresponding to the Tang Dynasty.

Puyang Tombs (濮阳墓地) (Contact: Juan Wang)

The Puyang Tombs, located in Puyang City, Henan Province, have yielded several burials through excavation. Faunal remains recovered from the site include a wide range of species such as domestic cattle, horse, felid, hare, canid, badger, deer, Chinese zokor, and pheasant. One sample (FS18) from Tomb M10 was analyzed in this study. Based on archaeological context, this individual is dated to as early as the late Tang Dynasty through the Five Dynasties and Ten Kingdoms period (618-970 CE)³.

Xitucheng City Site (西土城) (Contact: Zhuang Wu)

The Xitucheng City Site is at the western edge of Xitucheng Village, Erhao Bu Township, Kangbao County, Zhangjiakou City, Hebei Province. Archaeological evidence indicates that this site dates to the Jin Dynasty (1115-1234 CE), making it approximately 800 years old^{4,5}. Five domestic felid specimens (FS1, FS2, FS3, FS4, FS5) from this site were included in the present study.

Xinxiang Song Tomb (新乡宋墓) (Contact: Xianzhu Wu)

The Xinxiang Song Tomb, uncovered during construction of the new campus of the Henan Institute of Technology, functioned as a family burial ground during the Northern Song Dynasty (960-1127 CE). Excavation revealed four tombs of similar design, all imitating wooden structures with brick chambers. Each consisted of three sections: a burial shaft, an entrance, and a burial chamber. The shafts were oriented to the south, and the stratigraphy of the burial ground was relatively simple. In Tomb 2012XJZM4, alongside burial goods such as porcelain and copper coins, animal remains were recovered, including those of a domestic felid buried adjacent to the head of the tomb occupant⁶. One specimen from this context, FS6, was analyzed in the present study.

Zheng and Han City Site (郑韩故城) (Contact: Ziyi Li)

Zheng and Han City is in the heartland of Central China. During the Spring and Autumn and Warring States periods (770–221 BCE), it served successively as the capital of the

Zheng and Han states⁷. The felid bones analyzed in this study, however, were not from those early periods but from later contexts. Specifically, sample FS19 was unearthed from a tomb dating to the late Eastern Han period (25-220 CE) and radiocarbon dated to 200-380 cal CE, while samples FS20, FS21, FS22, and FS23 were recovered from pits dating between the Song and Yuan periods (960-1368 CE).

Zhengzhou Shang City Site (郑州商城) (Contact: Juan Wang)

Zhengzhou Shang City is in Henan Province, central China, and served as the capital during the Shang Dynasty^{8,9}. This site has remained in continuous use as a residential area through successive historical periods. In the present study, a felid bone (sample FS16) excavated from a pit at this site was analyzed. Radiocarbon dating places it within the Ming Dynasty, between 1413 and 1480 cal CE.

Shaliangzi City Site (沙梁子) (Contact: Chong Yu)

The Shaliangzi City Site, located in Hohhot, Inner Mongolia, covers an area of approximately 110,000 square meters. At its center, archaeologists uncovered a large, rammed-earth platform dating to the Han Dynasty. Artifacts recovered from the site mainly include architectural materials such as flat tiles, tube tiles, eave tiles, and square bricks with geometric patterns¹⁰. A felid skeleton (sample FS17) was discovered beneath the foundation of the granary. While the foundation itself was constructed during the Han Dynasty, radiocarbon dating indicates that the felid remains date to the Qing Dynasty (1671-1944 CE).

Quanhucun Site (泉护村) (Contact: Songmei Hu)

The Quanhucun Site is in the village of the same name in Hua County, Shaanxi Province, northwestern China. Archaeological remains at the site are attributed to the Middle to Late Yangshao Culture (4000-2700 BCE). The Yangshao people practiced agriculture, cultivating foxtail millet and common millet, and domesticated animals such as pigs and dogs. Previous studies indicate that leopard cats formed a commensal relationship with humans at this site¹¹⁻¹³. In this present study, one felid specimen (FS15), dated to approximately 5,400 years ago, was analyzed from the Quanhucun site¹¹⁻¹³.

Wuzhuang Guoliang Site (五庄果梁) (Contact: Songmei Hu)

The Wuzhuang Guoliang archaeological site is in Jingbian County, Shaanxi Province, northwestern China. Covering approximately 600,000 square meters across five mountain ridges, the site preserves cultural remains spanning the Neolithic Age to the Warring States period (c. 3000-221 BCE)^{12,14}. One felid sample (FS8), dating to the Neolithic period, was analyzed in this study.

Huoshiliang Site (火石梁) (Contact: Songmei Hu)

The Huoshiliang Site is in the desert interior of Yulin City, Shaanxi Province, northwestern China, and covered approximately 100,000 square meters. Numerous animal bones were recovered from the site. Preliminary analysis indicated that the primary cultural remains span from the late Longshan period of the Chinese Neolithic Age to the Erlitou period (c. 1800 BCE)¹⁵. One felid specimen (FS7) from this site was included in the present study.

Xinjie Site (新街) (Contact: Songmei Hu)

The Xinjie Site is a settlement on the loess plateau along the eastern bank of the Ba River, in the Wei River basin, Shaanxi Province, northwestern China. The site spans from the late Yangshao period to the Longshan period of the Chinese Neolithic Age (c. 3500-1800 BCE), with most remains dating to the late Yangshao period. The site extends approximately 600 meters east-west and 500 meters north-south, covering around 300,000 square meters¹⁶. One felid specimen (FS14) from the Longshan context was analyzed in this study.

Laoniupo Site (老牛坡) (Contact: Songmei Hu)

The Laoniupo Site is in the eastern suburb of Xi'an City, Shaanxi Province, China. The site was occupied by multiple archaeological cultures spanning the Neolithic period to the Shang Dynasty (c. 5000-1200 BCE)¹⁷. One felid specimen (FS10) from the Shang context was included in this study.

Han Chang'an City Site (汉长安城) (Contact: Songmei Hu)

The Han Chang'an City Site is in Weiyang District, Xi'an City, Shaanxi Province, China. Chang'an served as the capital of the Western Han Dynasty, built and expanded by Emperor Gaozu of Han and his successors on the foundation of the Qin Dynasty's Xingling Palace. In 2002, the Han Chang'an City Working Group from the Institute of Archaeology, Chinese Academy of Social Sciences, conducted excavations at the southwest corner of the city wall¹⁸. During this excavation, a felid tibia (sample FS24) was recovered. Radiocarbon dating places this sample between 44 cal BCE and 76 cal CE, confirming its attribution to the Han Dynasty.

Yangling Mausoleum (阳陵) (Contact: Songmei Hu)

Yangling Mausoleum is the joint burial tomb of Emperor Jing of the Western Han Dynasty, Liu Qi, and his queen (157-126 BCE), located in Xianyang City, Shaanxi Province, China. It is the easternmost of the emperor tombs on the Xianyang¹⁹. During excavations, a felid skeleton (sample FS13) was recovered from an external burial pit on the east side of the tomb mound within the mausoleum garden and analyzed in this study. Radiocarbon dating places the specimen between 168 and 48 cal BCE, confirming its attribution to the Western Han Dynasty.